

SECTION 11

ORDERING INFORMATION AND MECHANICAL DATA

This section provides pin assignments and package dimensions for the devices described in this manual.

11.1 PIN ASSIGNMENTS

Package	68000	68008	68010	68HC000	68HC001	68EC000
64-Pin Dual-In-Line	✓		✓	✓		
68-Terminal Pin Grid Array	✓		✓	✓	✓	
64-Lead Quad Pack						✓
68-Lead Quad Flat Pack	✓		✓	✓	✓	✓
52-Lead Quad		✓				
48-Pin Dual-In-Line		✓				

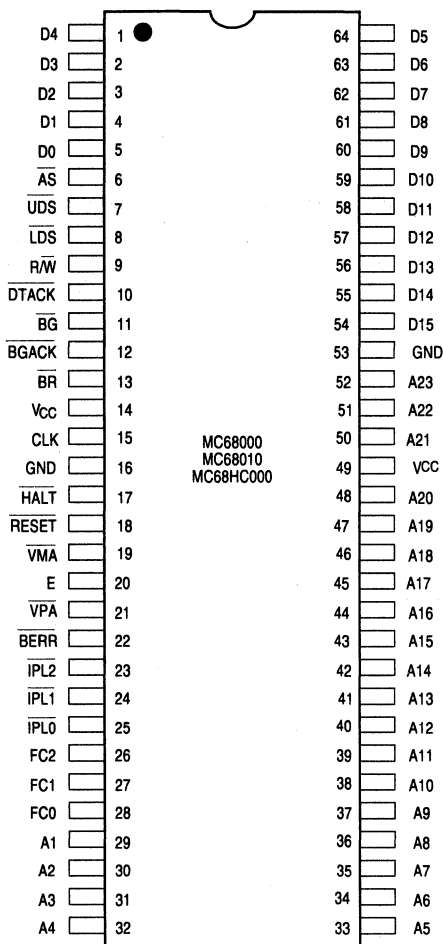
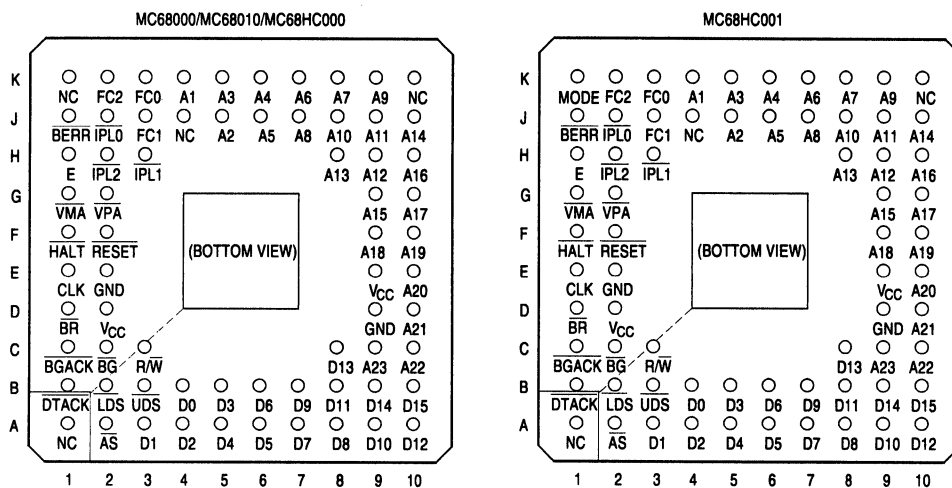


Figure 11-1. 64-Pin Dual In Line



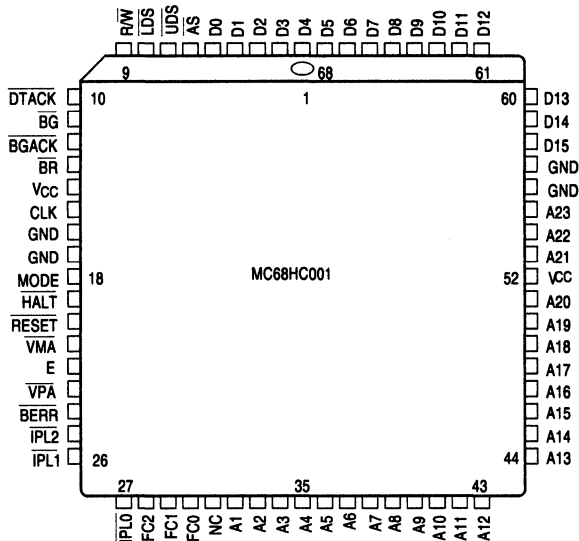


Figure 11-3. 68-Lead Quad Pack (2 of 2)

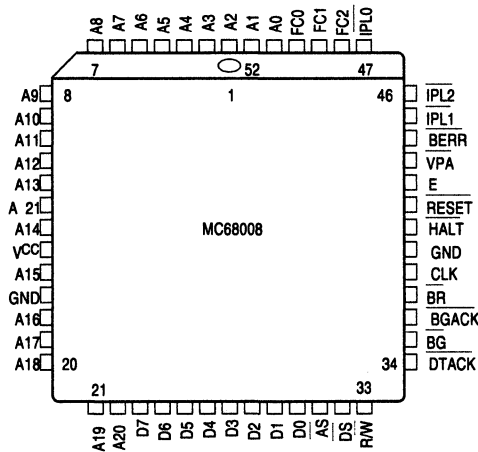


Figure 11-4. 52-Lead Quad Pack

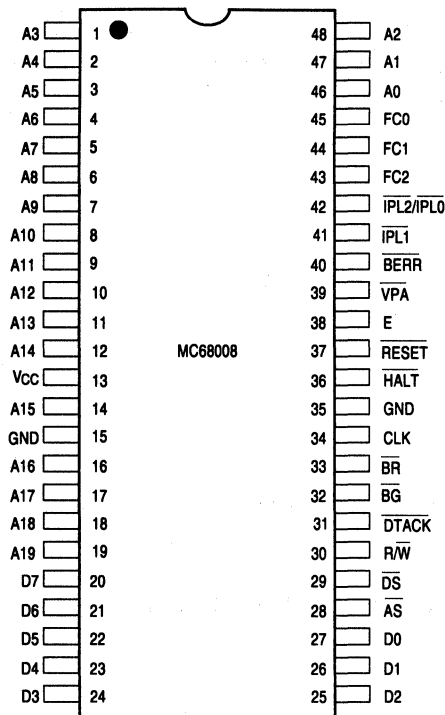


Figure 11-5. 48-Pin Dual In Line

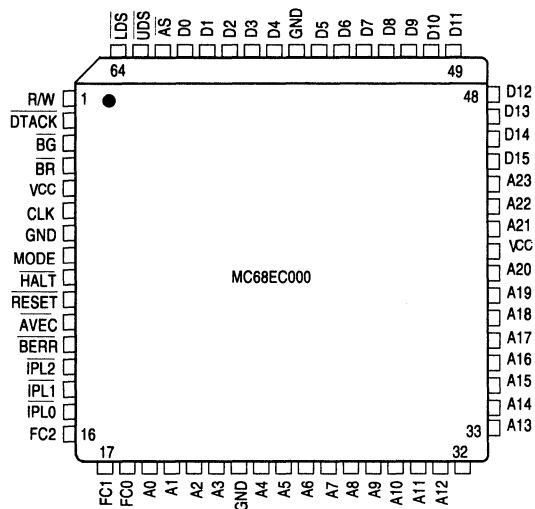
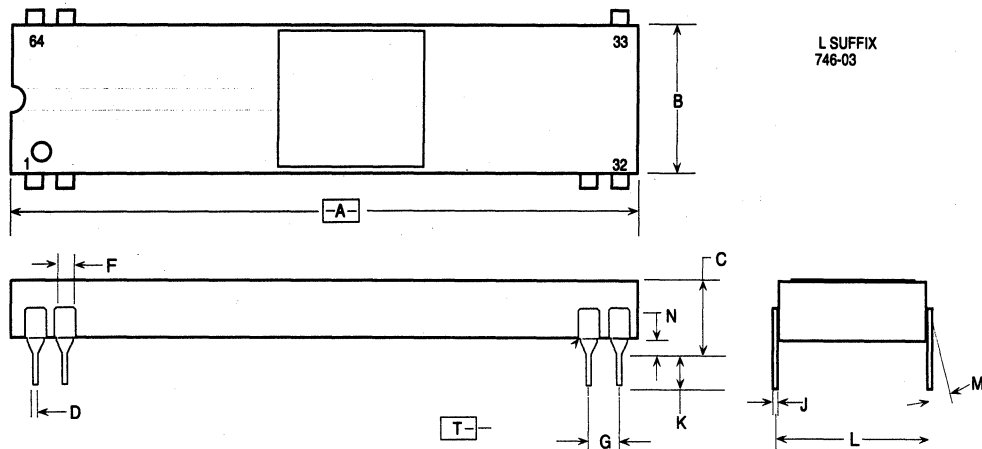


Figure 11-6. 64-Lead Quad Flat Pack

11.2 PACKAGE DIMENSIONS

Case Package	68000	68008	68010	68HC000	68HC001	68EC000
740-03 L Suffix		✓				
767-02 P Suffix		✓				
746-01 LC Suffix	✓		✓	✓		
754-01 R and P Suffix	✓		✓	✓		
765A-05 RC Suffix	✓		✓	✓	✓	
778-02 FN Suffix		✓				
779-02 FN Suffix				✓		✓
779-01 FN Suffix	✓		✓		✓	
847-01 FC Suffix				✓		
840B-01 FU Suffix						✓



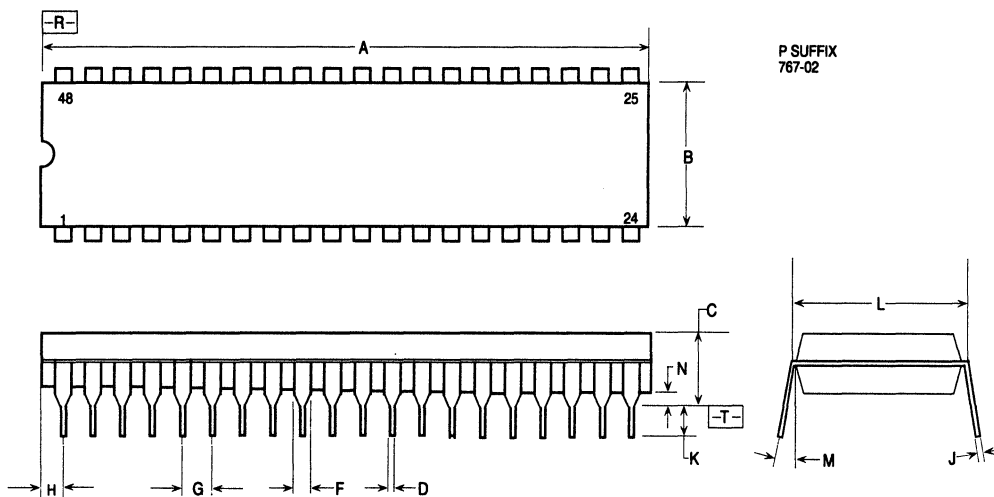
L SUFFIX
746-03

NOTES:

1. DIMENSION -A- IS DATUM.
2. POSITIONAL TOLERANCE FOR LEADS:
 $\text{M} \begin{smallmatrix} \text{P} \\ \text{O} \end{smallmatrix} 0.25 (0.010) \text{M} \text{T} \begin{smallmatrix} \text{A} \\ \text{M} \end{smallmatrix}$
3. -T- IS SEATING PLANE
4. DIMENSION "L" TO CENTER OF LEADS WHEN FORMED PARALLEL.
5. DIMENSIONING AND TOLERANCING PER ANSI Y14.5m, 1982.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	60.36	61.56	2.376	2.424
B	14.64	15.34	0.576	0.604
C	3.05	4.32	0.120	0.160
D	3.81	0.533	0.015	0.021
F	.762	1.397	0.030	0.055
G	2.54	BSC	0.100	BSC
J	0.204	0.330	0.008	0.013
K	2.54	4.19	0.100	0.165
L	15.24	BSC	0.600	BSC
M	0°	10°	0°	10°
N	1.016	1.524	0.040	0.060

Figure 11-7. Case 740-03—L Suffix

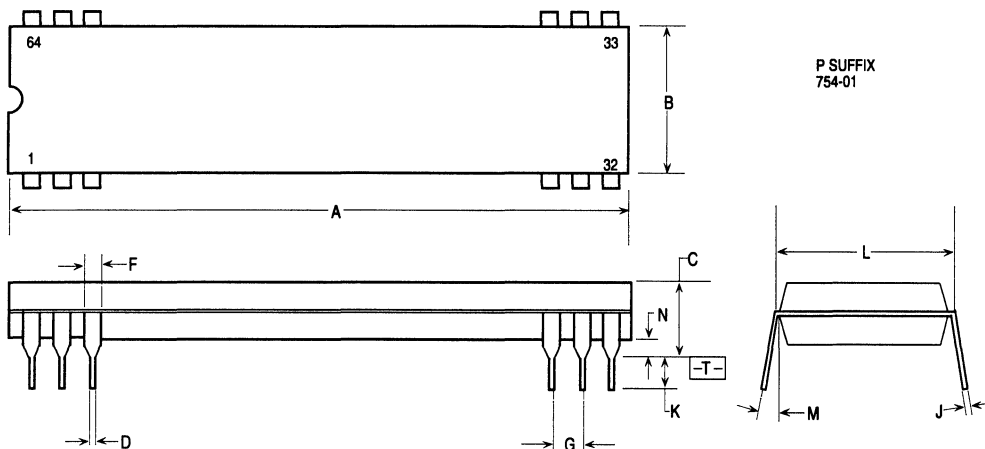


NOTES:

1. -R- IS END OF PACKAGE DATUM PLANE
-T- IS BOTH A DATUM AND SEATING PLANE
2. POSITIONAL TOLERANCE FOR LEADS 1 AND 48.
 $\boxed{\begin{smallmatrix} \oplus \\ 0.51 \end{smallmatrix} (0.020) \begin{smallmatrix} T \\ B \end{smallmatrix} \begin{smallmatrix} M \\ M \end{smallmatrix} R}$
 POSITIONAL TOLERANCE FOR LEAD PATTERN;
 $\boxed{\begin{smallmatrix} \oplus \\ 0.25 \end{smallmatrix} (0.020) \begin{smallmatrix} T \\ B \end{smallmatrix} \begin{smallmatrix} M \\ M \end{smallmatrix}}$
3. DIMENSION "A" AND "B" DOES NOT INCLUDE MOLD FLASH, MAXIMUM MOLD FLASH 0.25 (0.010).
4. DIMENSION "L" IS TO CENTER OF LEADS WHEN FORMED PARALLEL.
5. DIMENSIONING AND TOLERANCING PER ANSI Y14.5, 1982.
6. CONTROLLING DIMENSION: INCH.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	61.34	62.10	2.415	2.445
B	13.72	14.22	0.540	0.560
C	3.94	5.08	0.155	0.200
D	0.36	0.55	0.014	0.022
F	1.02	1.52	0.040	0.060
G	2.54	BSC	0.100	BSC
H	1.79	BSC	0.070	BSC
J	0.20	0.38	0.008	0.015
K	2.92	3.81	0.115	0.135
L	15.24	BSC	0.600	BSC
M	0°	15°	0°	15°
N	0.51	1.02	0.020	0.040

Figure 11-8. Case 767-02—P Suffix



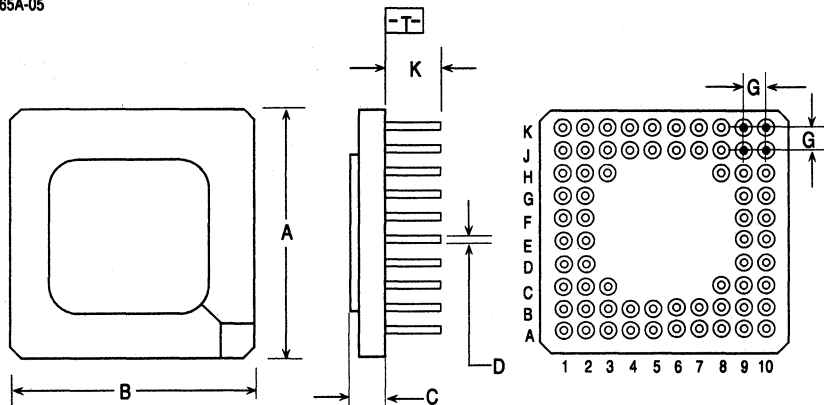
NOTES:

1. DIMENSIONS A AND B ARE DATUMS.
2. -T- IS SEATING PLANE.
3. POSITIONAL TOLERANCE FOR LEADS
(DIMENSION D):
 $\left(\begin{smallmatrix} \text{H} \\ \text{F} \end{smallmatrix} \right) \left(\begin{smallmatrix} \text{C} \\ \text{D} \end{smallmatrix} \right) 0.25 (0.010) \text{ (M) T (A) (M) B (M)}$
4. DIMENSION B DOES NOT INCLUDE MOLD FLASH.
5. DIMENSION L IS TO CENTER OF LEADS WHEN FORMED PARALLEL.
6. DIMENSIONING AND TOLERANCING PER ANSI Y14.5, 1982.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	81.16	81.91	3.195	3.225
B	20.17	20.57	0.790	0.810
C	4.83	5.84	0.190	0.230
D	0.33	0.53	0.013	0.021
F	1.27	1.77	0.050	0.070
G	2.54	BSC	0.100	BSC
J	0.20	0.38	0.008	0.015
K	3.05	3.55	0.120	0.140
L	22.86	BSC	0.9 00	BSC
M	0°	15°	0°	15°
N	0.51	1.02	0.020	0.040

Figure 11-10. Case 754-01—R and P Suffix

RC SUFFIX PACKAGE
CASE 765A-05



NOTES:

1. DIMENSIONS A AND B ARE DATUMS AND $\square$ -T- IS DATUM SURFACE.
2. POSITIONAL TOLERANCE FOR LEADS (68 PLACES).

ϕ	0.13 (0.005)	(M)	T	A	(S)	B	(S)
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3. DIMENSIONING AND TOLERANCING PER Y14.5M, 1982.
4. CONTROLLING DIMENSION: INCH.

DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	26.67	27.17	1.050	1.070
B	26.67	27.17	1.050	1.070
C	2.09	3.59	0.082	0.102
D	0.43	0.50	0.017	0.020
G	2.54 BSC		0.100 BSC	
K	4.32	4.82	0.170	0.190

R SUFFIX
PIN GRID ARRAY
WITH STANDOFF
(Dimensions essentially those of
Case 765A-05.)

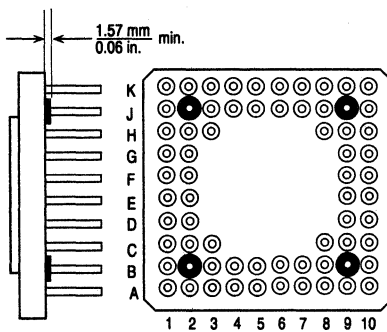


Figure 11-11. Case 765A-05—RC Suffix

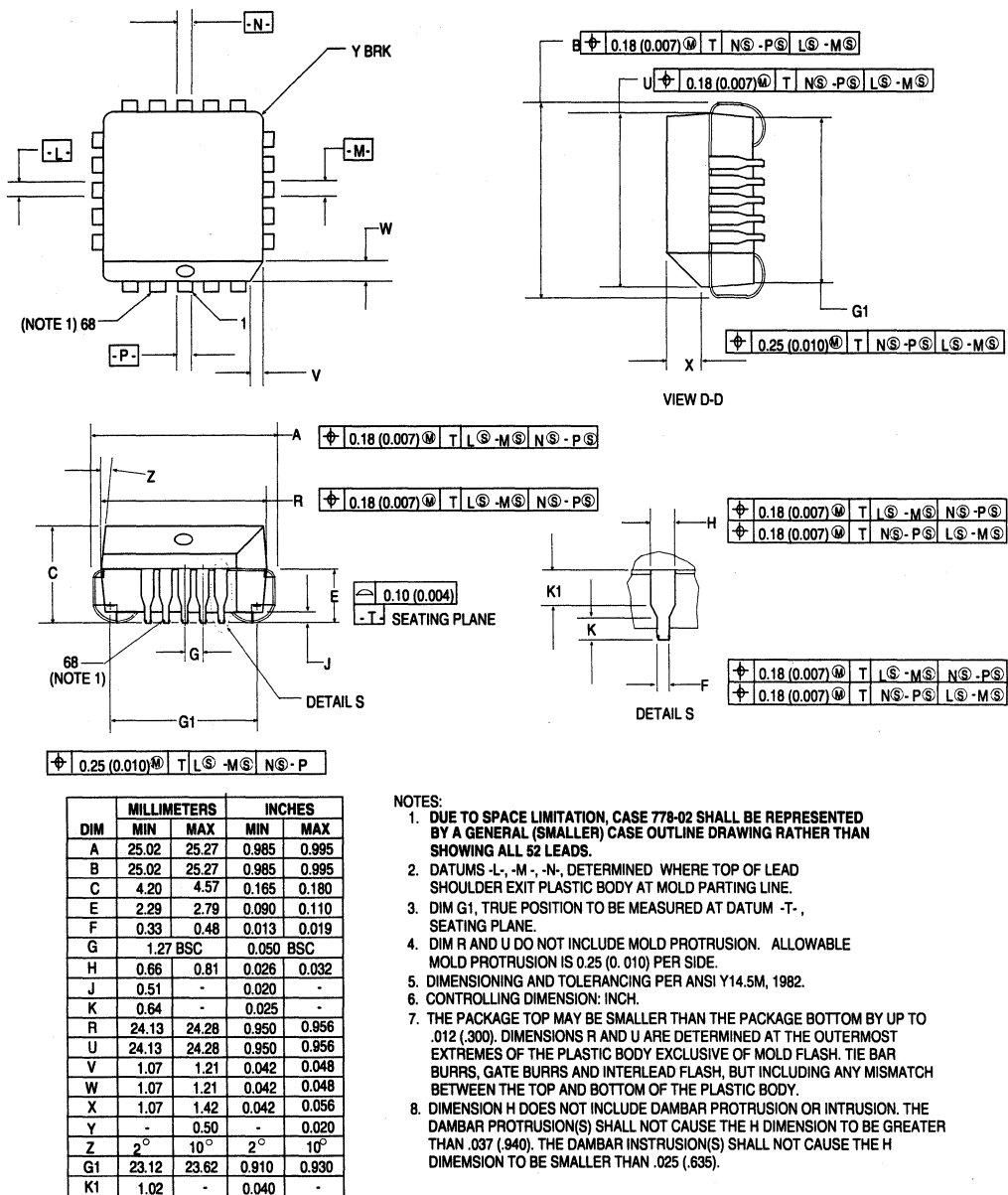
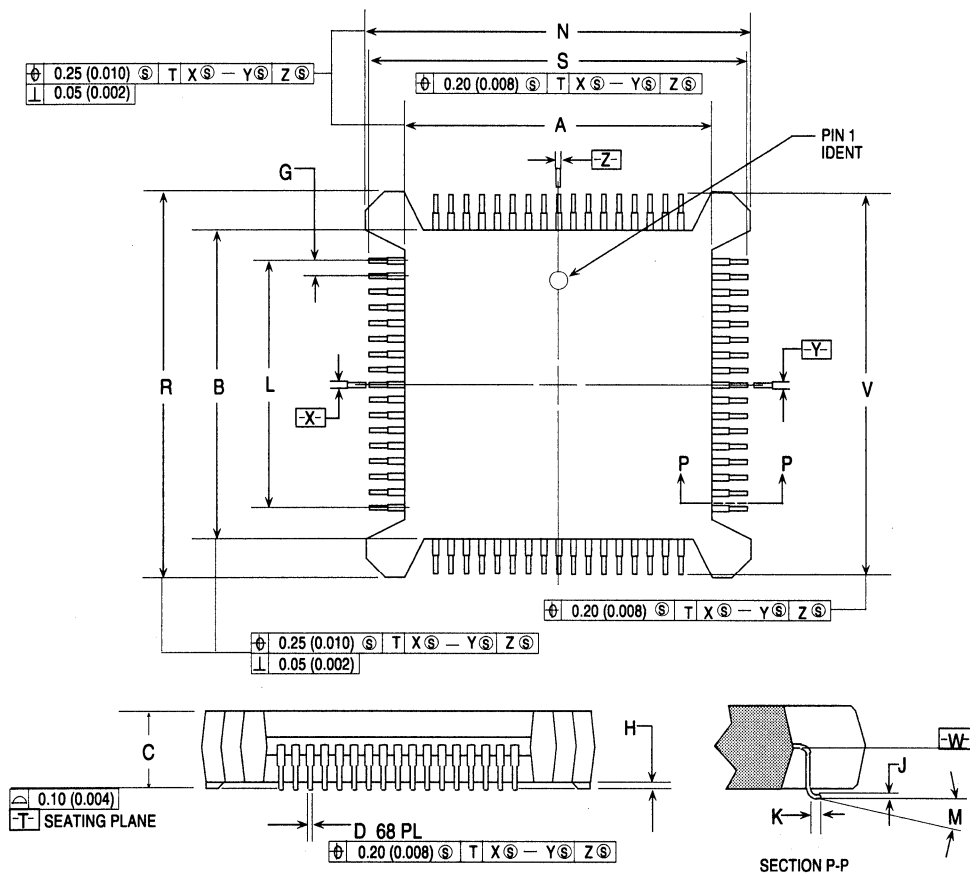


Figure 11-13. Case 779-02—FN Suffix

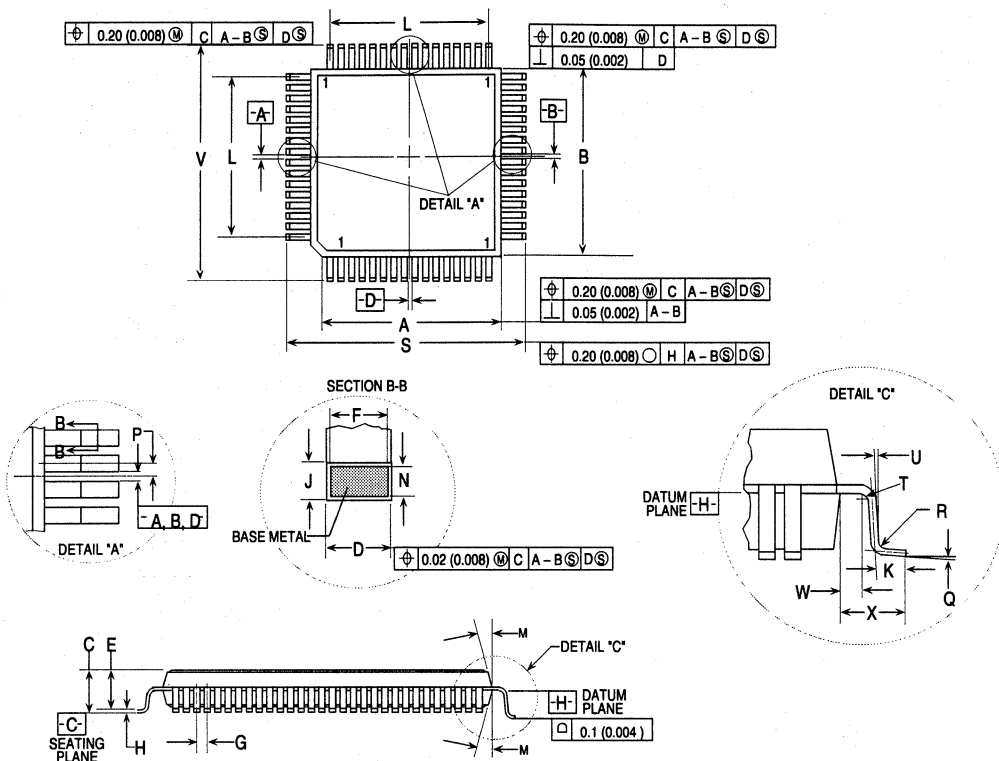


DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	13.90	14.04	0.547	0.553
B	13.90	14.04	0.547	0.553
C	2.14	3.04	0.084	0.120
D	0.21	0.30	0.008	0.012
G	0.64 BSC		0.025 BSC	
H	0.11	0.40	0.004	0.019
J	0.16	0.20	0.006	0.008
K	0.51	0.76	0.020	0.030
L	10.16 REF		0.400 REF	
M	0°	8°	0°	8°
N	17.71	17.85	0.697	0.703
R	17.71	17.85	0.697	0.703
S	17.15	17.39	0.675	0.685
V	17.15	17.39	0.675	0.685

NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCHES
3. DIM A, B, N, AND R DO NOT INCLUDE MOLD PROTRUSION. ALLOWABLE MOLD PROTRUSION FOR DIMENSIONS A AND B IS 0.25 (0.010), FOR DIMENSIONS N AND R IS 0.18 (0.007).
4. DATUM PLANE -W- IS LOCATED AT THE UNDERSIDE OF LEADS WHERE LEADS EXIT PACKAGE BODY.
5. DATUMS X-Y AND Z TO BE DETERMINED WHERE CENTER LEADS EXIT PACKAGE BODY AT DATUM -W-.
6. DIMENSIONS S AND V TO BE DETERMINED AT SEATING PLANE, DATUM -T-.
7. DIMENSIONS A, B, N AND R TO BE DETERMINED AT DATUM PLANE -W-.

Figure 11-14. Case 847-01—FC Suffix



NOTES:

1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: MILLIMETER. INCHES ARE IN ()
3. DATUM PLANE -H- IS LOCATED AT BOTTOM OF LEAD AND IS COINCIDENT WITH THE LEAD WHERE THE LEAD EXITS THE PLASTIC BODY AT THE BOTTOM OF THE PARTING LINE.
4. DATUMS -A-, -B-, AND -D- TO BE DETERMINED AT DATUM PLANE -H-.
5. DIMENSIONS S AND V TO BE DETERMINED AT SEATING PLANE -C-.
6. DIMENSIONS A AND B DO NOT INCLUDE MOLD PROTRUSION. ALLOWABLE PROTRUSION IS 0.25 (0.010) PER SIDE. DIMENSIONS A AND B DO INCLUDE MOLD MISMATCH AND ARE DETERMINED AT DATUM PLANE -H-.
7. DIMENSION D DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08 (0.003) TOTAL IN EXCESS OF THE D DIMENSION AT MAXIMUM MATERIAL CONDITION. DAMBAR CANNOT BE LOCATED ON THE LOWER RADIUS OR THE FOOT.

Figure 11-15. Case 840B-01—FU Suffix